

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

CO2 ASSESSMENT TOOL 2 – Architecture Design Assignment

Course Code:	CSA1004	Course Title:	Software Engineering
CO Assessed:	CO2	Assessment:	Assessment Tool 2 – Architecture Design Assignment
Weightage:	20%	Total Marks:	25
CO2 Statement:	Perform detailed requirements analysis, design scalable architectures, and prioritize features with a focus on cross-disciplinary skills		
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Date:		Duration:	60 minutes

Code of Conduct

I A.Sai Joshitha(192424220) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: A.Sai Joshitha

1. SYSTEM REQUIREMENT ANALYSIS

1.1 Introduction

The Online Movie Ticket Booking System is a software system that allows users to search for movies, view movie details, check theatres and show timings, select seats, make online payments, and receive booking confirmation. The system provides a convenient platform for customers to book movie tickets without visiting the theatre physically.

The system also allows administrators and theatre managers to manage movies, theatres, screens, show timings, seat availability, bookings, and transaction details.

The proposed system is designed using a suitable software architecture that provides scalability, security, performance, reliability, maintainability, and availability.

1.2 System Objectives

The main objectives of the Online Movie Ticket Booking System are:

1. To allow users to register and securely log in to the system.
2. To allow users to search and browse available movies.
3. To display movie information such as title, genre, language, duration, and rating.
4. To display available theatres and show timings.
5. To allow users to select available seats.
6. To prevent two users from booking the same seat simultaneously.
7. To provide secure online payment functionality.
8. To generate booking confirmation after successful payment.
9. To provide booking history to users.
10. To allow administrators to manage movies, theatres, screens, shows, seats, and bookings.
11. To maintain accurate booking and payment information.
12. To provide a reliable and user-friendly ticket booking service.

1.3 Functional Requirements

The major functional requirements are:

User Registration and Login

- Users should be able to create an account.
- Users should be able to log in using valid credentials.
- The system should authenticate users securely.

Movie Management

- Users should be able to view available movies.

- The system should display movie title, genre, language, duration, and other details.
- Administrators should be able to add, update, and remove movies.

Theatre and Show Management

- Users should be able to view available theatres.
- Users should be able to view show timings.
- Administrators should be able to manage theatres, screens, and show schedules.

Seat Selection

- Users should be able to view available seats.
- Users should be able to select one or more seats.
- The system should temporarily reserve selected seats during the booking process.
- Already booked seats should not be available for selection.

Booking

- Users should be able to confirm their selected movie, theatre, show, and seats.
- The system should generate a unique booking ID.
- Users should be able to view their booking history.

Payment

- The system should support online payment through a payment gateway.
- The payment status should be recorded.
- Booking confirmation should be generated only after successful payment.

Notification

- The system should provide booking confirmation to the user.
- The confirmation may contain movie, theatre, show time, seat numbers, and booking ID.

Administration

- Administrators should be able to manage movies.
- Administrators should be able to manage theatres and screens.
- Administrators should be able to manage show timings.
- Administrators should be able to monitor bookings and payments.

1.4 Non-Functional Requirements

Performance

The system should respond quickly to user requests such as movie search, show selection, and seat availability.

Scalability

The system should support an increasing number of users, theatres, movies, and bookings without major performance degradation.

Security

User credentials, personal information, and payment-related information must be protected from unauthorized access.

Reliability

The system should correctly process bookings and avoid duplicate or incorrect ticket reservations.

Availability

The booking service should remain available for users with minimum downtime.

Maintainability

The architecture should allow individual components to be modified or updated without affecting the entire system.

Usability

The system should provide a simple and user-friendly interface for searching movies and booking tickets.

2. SELECTION OF SUITABLE SOFTWARE ARCHITECTURE

2.1 Selected Architecture

For the Online Movie Ticket Booking System, a Three-Tier Layered Architecture is selected.

The three major layers are:

1. Presentation Layer
2. Business Logic Layer
3. Data Access Layer

The system also interacts with external services such as a payment gateway and notification service.

2.2 Architecture Structure

Presentation Layer

This layer provides the user interface through which customers and administrators interact with the system.

Main functions include:

- User registration and login
- Movie search
- Movie details
- Theatre and show selection
- Seat selection

- Booking details
- Payment interface
- Booking history
- Admin dashboard

Business Logic Layer

This layer contains the main business rules of the system.

It handles:

- User authentication
- Movie management
- Theatre management
- Show management
- Seat availability
- Booking processing
- Seat reservation
- Payment processing
- Booking validation
- Notification generation

Data Access Layer

This layer manages communication between the application and the database.

It performs operations such as:

- Storing user information
- Retrieving movie details
- Storing theatre and show information
- Managing seat information
- Storing booking records
- Storing payment details

Database

The database stores the persistent information of the system.

Important data includes:

- Users
- Movies
- Theatres
- Screens

- Seats
- Shows
- Bookings
- Payments

External Services

The system communicates with external services such as:

- Payment Gateway
- Email/SMS Notification Service

2.3 Justification for Selecting Three-Tier Architecture

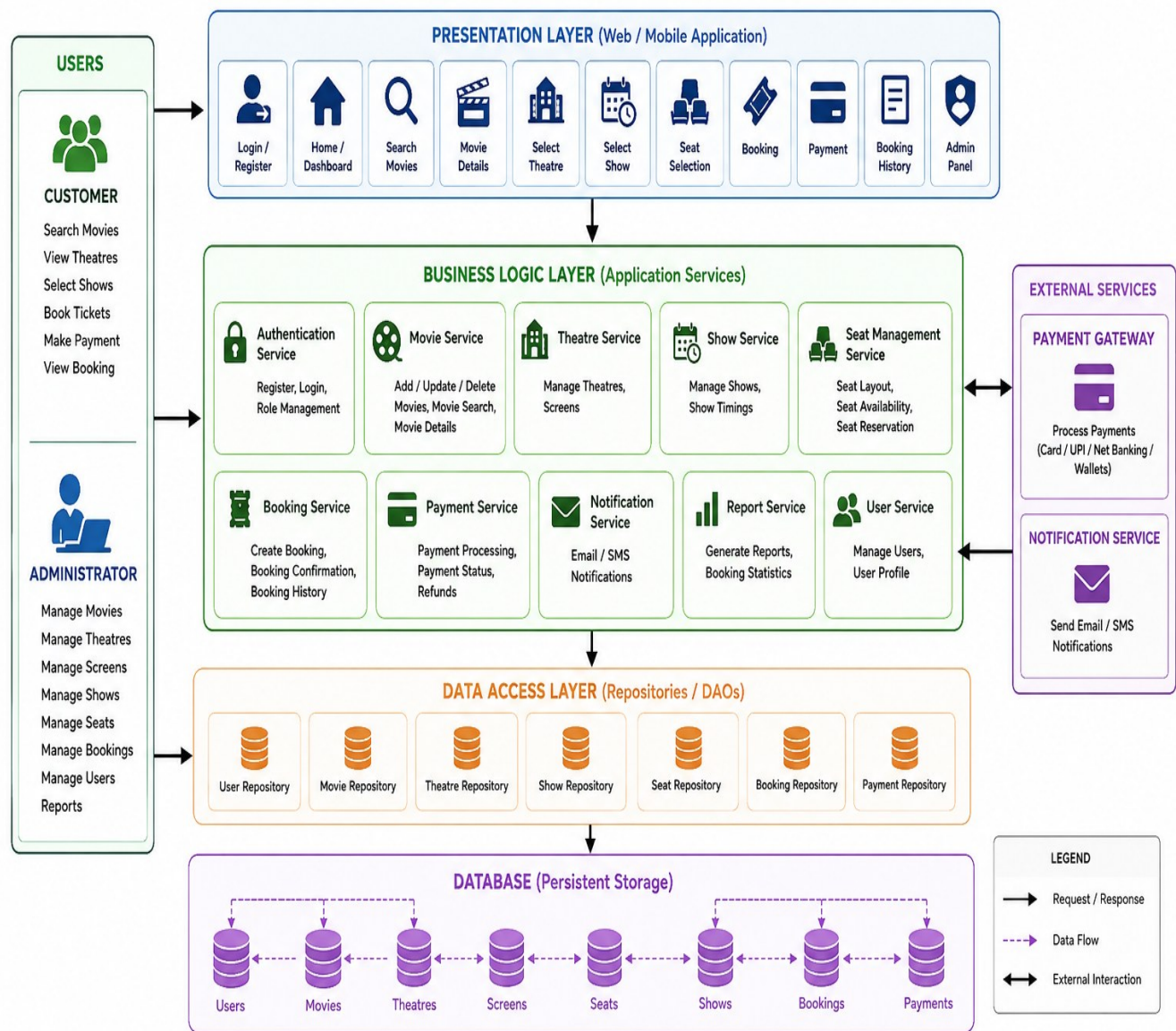
Three-Tier Architecture is suitable for the Online Movie Ticket Booking System because it separates the presentation, business logic, and data management responsibilities.

The main reasons are:

1. Separation of concerns: Each layer has a specific responsibility.
2. Maintainability: Changes in one layer can be made with minimal effect on other layers.
3. Security: Database access can be controlled through the business and data access layers.
4. Scalability: Individual layers can be improved or scaled based on system requirements.
5. Reusability: Business logic can be reused by different client interfaces.
6. Easy testing: Each layer can be tested independently.
7. Integration: External services such as payment gateways can be integrated through the business layer.

3. SOFTWARE ARCHITECTURE DIAGRAM

3.1 Architecture Diagram



Components to show in the diagram:

Users

- Customer
- Administrator

Presentation Layer

- Web/Mobile UI
- Login/Register
- Movie Search
- Show Selection
- Seat Selection
- Booking Interface
- Admin Dashboard

Business Logic Layer

- Authentication Service
- Movie Service
- Theatre & Show Service
- Seat Management Service
- Booking Service
- Payment Service
- Notification Service

Data Access Layer

- User Repository
- Movie Repository
- Theatre Repository
- Booking Repository
- Payment Repository

Database

- User Database
- Movie Database
- Theatre Database
- Seat Database
- Booking Database
- Payment Database

External Systems

- Payment Gateway
- Email/SMS Service

4. COMPONENTS AND THEIR INTERACTIONS

4.1 User Interface

The User Interface provides interaction between users and the system.

Responsibilities:

- Display available movies.
- Display theatres and show timings.
- Allow seat selection.
- Display booking information.
- Provide payment interface.
- Display booking confirmation.

4.2 Authentication Service

The Authentication Service manages user registration and login.

Responsibilities:

- Register users.
- Validate login credentials.
- Manage user sessions.
- Control access based on user roles.

4.3 Movie Service

The Movie Service manages movie-related information.

Responsibilities:

- Display movies.
- Search movies.
- View movie details.
- Add, update, and remove movies for administrators.

4.4 Theatre and Show Service

This component manages theatre, screen, and show information.

Responsibilities:

- Display theatres.
- Display screens.
- Display show timings.

- Manage movie-show relationships.

4.5 Seat Management Service

This component manages seat availability.

Responsibilities:

- Display available seats.
- Allow users to select seats.
- Temporarily reserve seats during booking.
- Release seats when a booking fails or expires.
- Prevent duplicate seat booking.

4.6 Booking Service

The Booking Service manages ticket booking.

Responsibilities:

- Validate selected movie, theatre, show, and seats.
- Create booking records.
- Generate booking ID.
- Confirm booking after successful payment.
- Maintain booking history.

4.7 Payment Service

The Payment Service handles payment-related operations.

Responsibilities:

- Initiate payment.
- Communicate with the payment gateway.
- Receive payment status.
- Update transaction information.
- Inform the Booking Service about successful or failed payments.

4.8 Notification Service

The Notification Service sends booking-related notifications.

Responsibilities:

- Send booking confirmation.
- Send ticket details.
- Notify users about booking status.

4.9 Data Access Layer

The Data Access Layer communicates with the database.

Responsibilities:

- Insert data.
- Retrieve data.
- Update data.
- Delete data.
- Provide controlled access to stored information.

4.10 Database

The database stores all persistent system information.

Major entities include:

- User
- Movie
- Theatre
- Screen
- Seat
- Show
- Booking
- Payment

4.11 COMPONENT INTERACTION**The overall booking workflow is as follows:**

1. The user opens the Online Movie Ticket Booking System.
2. The user registers or logs into the system.
3. The user searches for a movie.
4. The system retrieves movie information from the database.
5. The user selects a theatre and show timing.
6. The system displays the available seats.
7. The user selects the required seats.
8. The Seat Management Service temporarily reserves the selected seats.
9. The user confirms the booking.
10. The Booking Service creates a pending booking.
11. The Payment Service sends the payment request to the external Payment Gateway.
12. The Payment Gateway processes the payment.
13. If the payment is successful, the booking is confirmed.
14. The selected seats are marked as booked.

15. The booking details are stored in the database.
16. The Notification Service sends confirmation to the user.
17. The user receives the booking ID and ticket details.\

5. QUALITY ATTRIBUTE ANALYSIS

5.1 Scalability

The architecture supports scalability by separating the system into independent layers.

The application layer can be scaled when the number of users increases. Database optimization and caching can also be introduced to handle a large number of movie searches and seat availability requests. For example, during weekends or festival periods, many users may attempt to book tickets simultaneously. The architecture can be scaled to handle the increased workload.

5.2 Security

Security is important because the system handles user accounts and payment transactions.

Security can be provided through:

- Secure authentication
- Password encryption/hashing
- Role-based access control
- HTTPS communication
- Input validation
- Secure database access
- Secure payment gateway integration

Administrators should have higher privileges than normal users.

5.3 Performance

The architecture improves performance by separating responsibilities between layers.

Performance can be improved using:

- Database indexing
- Caching frequently requested movie information
- Efficient database queries
- Optimized APIs
- Load balancing when required

Seat availability and movie search operations should return results quickly.

5.4 Reliability

Reliability means that the system should perform booking operations correctly.

The architecture provides reliability through:

- Transaction management
- Database consistency
- Payment verification
- Booking validation
- Prevention of duplicate seat booking
- Proper error handling

If payment fails, the booking should not be confirmed and temporarily reserved seats should be released.

5.5 Maintainability

The layered structure makes the system easier to maintain.

For example:

- User interface changes can be made in the Presentation Layer.
- Booking rules can be modified in the Business Logic Layer.
- Database-related changes can be handled in the Data Access Layer.

This reduces the effect of changes on other parts of the system.

5.6 Availability

The system should be available whenever users want to search for movies or book tickets.

Availability can be improved using:

- Backup systems
- Database backup and recovery
- Server redundancy
- Load balancing
- Monitoring
- Fault handling

These mechanisms help reduce system downtime.

6. ARCHITECTURAL DECISIONS AND JUSTIFICATION

6.1 Architectural Decision

The Three-Tier Layered Architecture was selected because the Online Movie Ticket Booking System contains different responsibilities that can be clearly separated.

The Presentation Layer handles user interaction, the Business Logic Layer handles system **operations**, and the Data Access Layer manages database communication.

6.2 Advantages

The selected architecture provides the following advantages:

1. Clear separation of responsibilities.

2. Easier maintenance.
3. Better security.
4. Easier testing.
5. Improved scalability.
6. Reusability of business logic.
7. Easier integration with external services.
8. Better organization of the system.

6.3 Trade-Offs

Although Three-Tier Architecture provides many benefits, there are some trade-offs.

Advantages

- Simple to understand.
- Easy to develop and maintain.
- Suitable for medium and large applications.
- Provides good separation of concerns.
- Easier debugging and testing.

Limitations

- Communication between layers can introduce some overhead.
- Very large systems may require additional architectural approaches.
- Scaling the complete application may sometimes be less flexible than independently deployable microservices.

For the proposed Online Movie Ticket Booking System, the benefits are greater than these limitations.

6.4 Future Enhancements

The system can be enhanced in the future by adding:

1. Mobile application support.
2. AI-based movie recommendations.
3. Multiple payment methods.
4. Digital wallet integration.
5. Promotional coupons and discounts.
6. Loyalty and reward programs.
7. Real-time seat availability.
8. Push notifications.
9. Multi-language support.
10. Cloud deployment.

11. Analytics and reporting.

12. Integration with multiple theatre chains.

If the system becomes very large, selected services can later be converted into independent microservices.

7. SYSTEM WORKFLOW SUMMARY

The overall workflow of the Online Movie Ticket Booking System is:

User Login → Movie Search → Select Movie → Select Theatre → Select Show → View Seats → Select Seats → Confirm Booking → Online Payment → Payment Verification → Booking Confirmation → Notification

This workflow ensures that the user can complete the ticket booking process in a systematic and secure manner.

8. CONCLUSION

The Online Movie Ticket Booking System requires an architecture that can provide security, scalability, performance, reliability, availability, and maintainability. Based on the system requirements, a Three-Tier Layered Architecture is selected as the suitable architectural style.

The Presentation Layer provides user interaction, the Business Logic Layer manages booking and business operations, and the Data Access Layer communicates with the database. External services such as payment gateways and notification services are integrated with the system.

The proposed architecture provides clear separation of responsibilities and supports future enhancements. It is therefore suitable for designing a reliable and maintainable Online Movie Ticket Booking System.

Rubrics for Calculation

Evaluation Criteria	Excellent (4 Marks)	Good (3 Marks)	Satisfactory (2 Marks)	Needs Improvement (1 Mark)
System Requirement Analysis (30%)	Thoroughly analyzes the problem statement, accurately identifies functional and non-functional requirements, and clearly explains quality attributes influencing the architecture.	Analyzes most requirements with minor omissions or limited explanation.	Identifies only basic requirements with partial analysis.	Requirement analysis is incomplete, inaccurate, or lacks clarity.
Selection of Software Architecture (25%)	Selects the most appropriate architectural style with strong technical justification aligned to system requirements.	Selects a suitable architecture with reasonable justification.	Architecture is partially suitable with limited justification.	Inappropriate architecture selected or justification is missing.
Architecture Diagram and Component Design (20%)	Produces a clear, complete, and well-structured architecture diagram with all major components, interfaces, and interactions accurately represented.	Diagram is mostly complete with minor omissions or notation issues.	Diagram includes only essential components and lacks sufficient detail.	Diagram is incomplete, unclear, or technically incorrect.
Component Interaction and Quality Attribute Analysis (15%)	Clearly explains component responsibilities, interactions, and thoroughly discusses scalability, security, performance, reliability, maintainability, and availability.	Explains most components and quality attributes with minor gaps.	Provides limited explanation of interactions and addresses only some quality attributes.	Component interactions and quality attributes are poorly explained or missing.

Architectural Justification and Technical Presentation (10%)	Provides strong justification for architectural decisions, discusses trade-offs and future enhancements, and presents a well-organized, professional report.	Provides adequate justification with minor gaps; report is well organized.	Limited justification with minimal discussion of trade-offs or future scope; report needs improvement.	Justification is weak or absent; report lacks organization and technical clarity.
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Criteria	Mark
System Requirement Analysis (30%)	/5
Selection of Software Architecture (25%)	/5
Architecture Diagram and Component Design (20%)	/5
Component Interaction and Quality Attribute Analysis (15%)	/5
Architectural Justification and Technical Presentation (10%)	/5
TOTAL	/25